Ibrahim Tariq

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IbrahimT04

Education

McMaster University — **Hamilton, ON**
BEng in Software Engineering & Coop — *2022–2027*

- **GPA:** 9.8/12.0; **Dean's Honor List:** Faculty of Engineering Achievement Award

Experience

Environment and Climate Change Canada – WMS East — **Burlington, ON**
Software Engineering Intern — *May 2025–August 2026*

- Refactored a legacy **hydrologic model** into a modular, data-driven **batch sequencing architecture** with replaceable components, reducing code size by 25%, improving plan production productivity by 6%, and lowering simulation runtime by 3%.
- Built Bash tooling around **PyLint** to automate pull request checks, summarizing style violations, refactor candidates, warnings, and errors into a readiness score and merge gate indicator.
- Produced onboarding documentation and delivered co-development presentations on coding practices and version control to scientists and engineers across Canadian and U.S. federal agencies.

DataAnnotation — **Toronto, ON**
Swift Developer — *May 2024–Aug 2024*

- Generated and assessed data used to train ML models, interacting with **over 10 LLMs** and evaluating **1000+ prompts**.
- Improved machine learning models' ability to program data structures in Python, Java, C++, TypeScript, and Swift.

Projects

RayTraceMax (OpenGL 4.6, GLFW, Python): Built a GPU compute shader pipeline for physically based rendering: Monte Carlo path tracing with Multiple Importance Sampling and Next-Event Estimation. Implemented microfacet Cook–Torrance BRDFs (GGX/Smith, Schlick Fresnel) with detailed materials (metallic/roughness, IOR, clearcoat, transmission), HDR IBL, and SSBO-backed geometry.

GraphiXion (OpenGL, GLSL, Python): Developed a 3D rendering engine with customizable shaders, lighting, texture mapping, Parallax Occlusion Mapping, and Soft Shadow Mapping. Implemented Physically Based Rendering techniques. Used a Tangent-Bitangent-Normal (TBN) matrix for normal map transformations, enhancing object realism.

Block Shooter (Python, PyGame, NumPy): Created an old-fashioned FPS with adjustable level design, enemy types, and scoring. Implemented ray casting, anti-aliasing (Xiaolin Wu algorithm), DDA line drawing, and frame tracking to ensure a 60Hz refresh rate.

Clarify (OpenCV, C++): Developed an open-source image processing application with efficient algorithms for image enhancement, manipulation, and analysis. Implemented image sharpening, blurring, noise reduction, edge detection, and basic image filtering techniques.

Extracurriculars

McMaster Sumobot Club — **Hamilton, ON**
Tournament-style, one-on-one matches with autonomous fighting robots — *Oct 2024–Present*

McMaster Game Development Club — **Hamilton, ON**
Game jams with: Unity, Unreal, Godot & Blender — *Sep 2024–Present*

Mississauga Secondary School — **Mississauga, ON**
Academic Tutor (Physics & Mathematics) — *Sep 2021–Jun 2022*

Skills

Languages: Python, C++, C, GLSL, Java, SQL, Swift, TypeScript

Frameworks & APIs: OpenGL, SDL3, PyGame, OpenCV, Unreal Engine 5, Godot, Unity, Blender

Tools: Unix, Vim, Git, CMake, Maven, Quartus, MATLAB, Arduino, MS Office